

seven sites across the United States and spanning a wide range of environmental conditions. At all sites, the RMS difference between the model-calculated WBGT and the measurements was less than 1°C. When the instrumentation was operating correctly the WBGT model predictions exhibited a bias of 0.4°C or less and an RMSE of less than 0.6°C; 91–100% of the WBGT predictions were within 1°C of the measured value. This is true even when using 2-m wind speed values estimated from wind speeds measured at other heights.

Local spatial variations in measured WBGT are shown to be comparable to or greater than the differences between model predictions and measurements. This result suggests it may be useful to calculate WBGT for multiple locations when the necessary meteorological input data are available to achieve maximum accuracy. It also suggests that calculating WBGT may be as useful as measuring WBGT with a device that requires careful, time-consuming attention.

Fortran and C-language programs that implement the Argonne WBGT model are available from the authors.

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